



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-4

Crypt Arithmetic Problem

Aim

To write a Python program to solve a Crypt Arithmetic Problem.

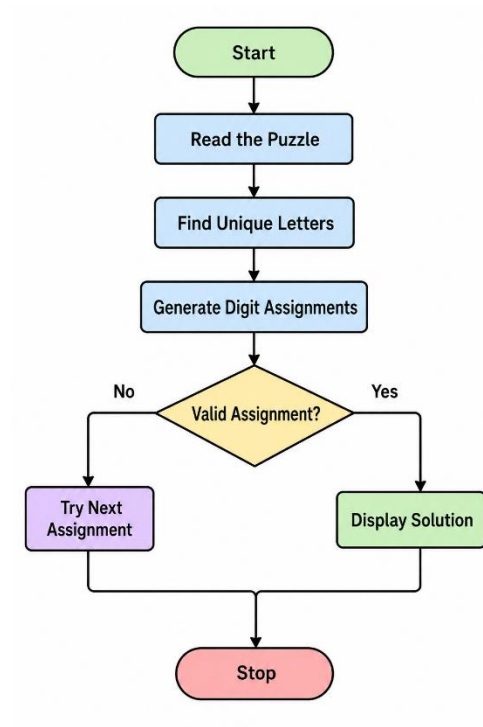
Objective

To solve a crypt arithmetic puzzle by assigning unique digits to letters.

Algorithm

1. Start.
2. Read the words in the crypt arithmetic puzzle.
3. Identify all unique letters.
4. Generate all possible digit assignments.
5. Check that no leading letter is assigned 0.
6. Verify whether the arithmetic equation is satisfied.
7. If a valid assignment is found, display the solution.
8. Stop.

Flowchart



Code

```
from itertools import permutations
```

```
letters = "SENDMORY"
```

```
digits = range(10)
```

```
for p in permutations(digits, len(letters)):
```

```
    d = dict(zip(letters, p))
```

```
    if d['S'] == 0 or d['M'] == 0:
```

```
        continue
```

```
    send = 1000*d['S'] + 100*d['E'] + 10*d['N'] + d['D']
```

```
    more = 1000*d['M'] + 100*d['O'] + 10*d['R'] + d['E']
```

```
    money = 10000*d['M'] + 1000*d['O'] + 100*d['N'] + 10*d['E'] + d['Y']
```

```

if send + more == money:

    print("SEND =", send)

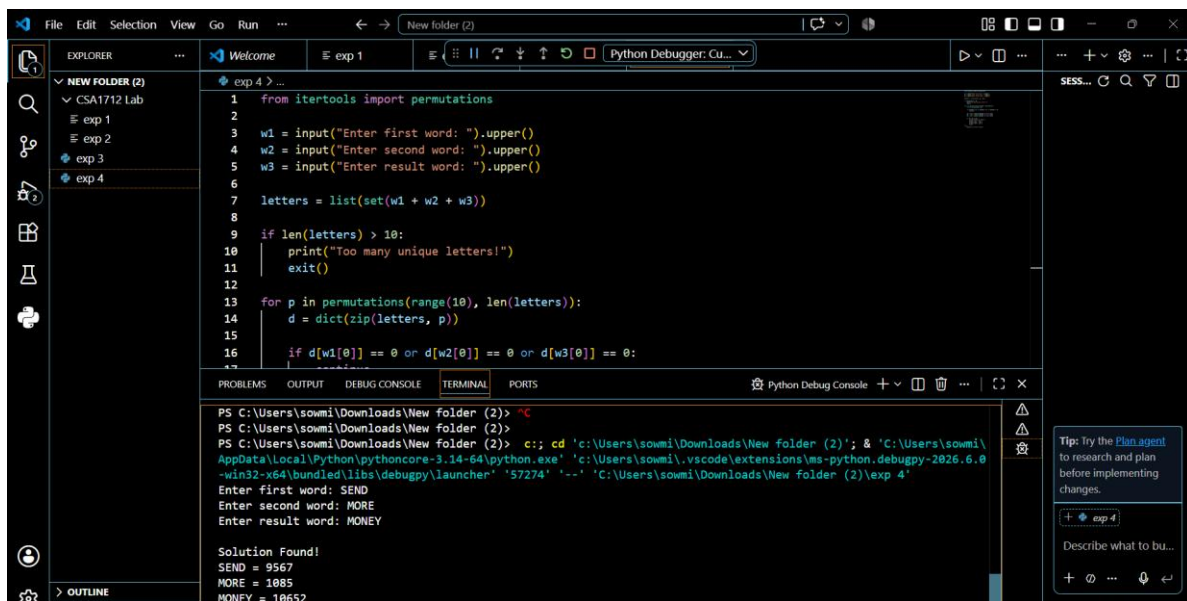
    print("MORE =", more)

    print("MONEY =", money)

    break

```

Output



The screenshot shows a VS Code editor with a Python script in the main editor and its execution output in the terminal. The script is a cryptarithm solver that takes three words (SEND, MORE, MONEY) as input and finds a valid digit assignment for the letters. The terminal output shows the program running successfully and finding the solution.

```

exp 4 > ...
1 from itertools import permutations
2
3 w1 = input("Enter first word: ").upper()
4 w2 = input("Enter second word: ").upper()
5 w3 = input("Enter result word: ").upper()
6
7 letters = list(set(w1 + w2 + w3))
8
9 if len(letters) > 10:
10     print("Too many unique letters!")
11     exit()
12
13 for p in permutations(range(10), len(letters)):
14     d = dict(zip(letters, p))
15
16     if d[w1[0]] == 0 or d[w2[0]] == 0 or d[w3[0]] == 0:
17         continue
18
19     if int(d[w1] * 10 + d[w2]) == int(d[w3]):
20         print("Solution Found!")
21         print("SEND = ", d[w1])
22         print("MORE = ", d[w2])
23         print("MONEY = ", d[w3])
24         break

```

Terminal Output:

```

PS C:\Users\sowmi\Downloads\New folder (2)> cd 'c:\Users\sowmi\Downloads\New folder (2)'; & 'C:\Users\sowmi\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\sowmi\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '57274' '-...' 'C:\Users\sowmi\Downloads\New folder (2)\exp 4'
Enter first word: SEND
Enter second word: MORE
Enter result word: MONEY

Solution Found!
SEND = 9567
MORE = 1085
MONEY = 10652

```

Result

The crypt arithmetic problem was solved successfully.

Conclusion

The program correctly finds a valid digit assignment for the given puzzle.